

Exact Coexistence of Justified Envy-Freeness and Pareto Optimality

A Correction and Partial Progress on Open Problem W4414541699

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Status. This note gives *partial progress*, not a complete solution to the stated algorithmic open problem. It proves that every finite two-sided cardinal matching instance has an *exactly* justified-envy-free and Pareto-optimal fractional matching, and that one can choose such a matching rationally. It also gives a polynomial-time, instance-dependent approximation guarantee that is always at most n . What remains open is a polynomial-time algorithm with a dimension-free constant approximation factor (or an exact polynomial-time algorithm).

Abstract

A proposed negative resolution of Open Problem W4414541699 claimed that, for every constant C , some two-sided cardinal matching instance has no Pareto-optimal C -approximately justified-envy-free lottery. That conclusion cannot hold. Tröbst and Vazirani proved that for every additive tolerance $\eta > 0$ there is a Pareto-optimal allocation whose justified-envy violations are at most η . We observe that, for a polytope with linear utilities, the set of Pareto-optimal points is a finite union of faces and is therefore closed. Passing to the limit in their construction consequently gives an allocation that is simultaneously exactly justified-envy-free and Pareto-optimal. A face argument further yields a rational such allocation. This rules out every nonexistence construction, including constructions using zero utilities. We also give a simple polynomial-time algorithm returning a Pareto-optimal κ -JEF allocation, where $\kappa \leq n$ is computed by a max–min linear program. Thus the existential issue is settled exactly, while the dimension-free polynomial-time approximation question remains open.

1 Introduction

Tröbst and Vazirani [2] ask whether there is a polynomial-time algorithm that computes an α -approximately justified-envy-free and Pareto-optimal lottery in every two-sided cardinal-utility matching market, for some universal constant α . The question is recorded as Open Problem W4414541699 at https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4414541699_p0&n=31&page=3.

The submitted manuscript that motivated this note attempted to answer the question negatively by constructing a “Pareto-simplex selector” on which every Pareto-optimal allocation has large multiplicative justified envy. The construction cannot be repaired as a nonexistence proof. The obstruction is not merely a missing gadget or a perturbation issue: every finite instance admits an allocation that is both exactly JEF and Pareto optimal.

The argument is short but consequential. Tröbst and Vazirani already construct, for every $\eta > 0$, a Pareto-optimal allocation with additive JEF error at most η ; see their Theorem 43 and Lemmas 44–45. Their proof then takes a convergent subsequence and records only weak Pareto optimality of the limit. In the present finite-dimensional polyhedral setting, however, the Pareto-optimal set is closed. Hence the limit is Pareto optimal in the strong sense as well.

Our contributions are as follows.

1. We give a self-contained scalarization proof that the Pareto-optimal set of a polytope under finitely many linear utilities is a finite union of faces, and hence compact.
2. Combining this fact with the additive approximation sequence of Tröbst and Vazirani proves exact coexistence of JEF and Pareto optimality. We also prove that a rational JEF+PO allocation exists and, more specifically, that some vertex of the JEF polytope is globally Pareto optimal.
3. We show directly that the proposed Pareto-simplex selector contradicts this theorem by a uniform positive gap.
4. We give a polynomial-time two-stage linear-programming algorithm that returns a Pareto-optimal κ -JEF allocation, where $\kappa \leq n$ and κ can be substantially smaller on a given instance.

The first three items settle the existential version of the question exactly. They do not identify the supporting Pareto face in polynomial time, so they do not resolve the algorithmic constant-factor question.

2 Model and definitions

Let A and B be two sets of agents with $|A| = |B| = n$. Agent $i \in A$ has a nonnegative cardinal utility u_{ij} for being matched to $j \in B$, and agent $j \in B$ has a nonnegative cardinal utility w_{ji} for being matched to $i \in A$. Unless stated otherwise, all data are rational.

A fractional perfect matching is a bistochastic matrix $x = (x_{ij})_{i \in A, j \in B}$ belonging to

$$\mathcal{X} := \left\{ x \in \mathbb{R}_+^{n \times n} : \sum_{j \in B} x_{ij} = 1 \ \forall i \in A, \quad \sum_{i \in A} x_{ij} = 1 \ \forall j \in B \right\}.$$

The expected utilities are

$$U_i(x) := \sum_{j \in B} u_{ij} x_{ij}, \quad W_j(x) := \sum_{i \in A} w_{ji} x_{ij}.$$

For distinct agents $i, i' \in A$, define the justified-envy benchmark

$$J_{i \leftarrow i'}^A(x) := \sum_{j \in B: w_{ji} \geq w_{ji'}} u_{ij} x_{ij}.$$

Thus i evaluates the portion of i' 's lottery consisting of partners who weakly prefer i to i' . Symmetrically, for distinct $j, j' \in B$, set

$$J_{j \leftarrow j'}^B(x) := \sum_{i \in A: u_{ij} \geq u_{ij'}} w_{ji} x_{ij}.$$

Definition 2.1 (JEF and multiplicative JEF). An allocation $x \in \mathcal{X}$ is *justified-envy-free* if

$$U_i(x) \geq J_{i \leftarrow i'}^A(x) \quad \forall i \neq i' \in A, \quad W_j(x) \geq J_{j \leftarrow j'}^B(x) \quad \forall j \neq j' \in B.$$

For $C \geq 1$, it is C -JEF if the right-hand sides may be multiplied by $1/C$, equivalently if

$$J_{i \leftarrow i'}^A(x) \leq C U_i(x), \quad J_{j \leftarrow j'}^B(x) \leq C W_j(x)$$

for all relevant pairs.

Definition 2.2 (Pareto optimality). An allocation $y \in \mathcal{X}$ Pareto dominates $x \in \mathcal{X}$ if

$$U_i(y) \geq U_i(x) \quad \forall i \in A, \quad W_j(y) \geq W_j(x) \quad \forall j \in B,$$

and at least one inequality is strict. An allocation is Pareto optimal if it is not Pareto dominated by another allocation in $\mathcal{X}$.

This is the stronger of the two standard Pareto notions. Weak Pareto optimality only rules out allocations that strictly improve every agent.

3 The additive Pareto approximants

We use the following consequence of the double-indexed-price equilibrium argument in Tröbst and Vazirani [2, Theorem 43 and Lemmas 44–45].

Proposition 3.1 (Tröbst–Vazirani additive approximation). *For every finite two-sided cardinal matching instance and every $\eta > 0$, there exists a Pareto-optimal allocation $x^\eta \in \mathcal{X}$ such that*

$$\begin{aligned} U_i(x^\eta) &\geq J_{i \leftarrow i'}^A(x^\eta) - \eta & \forall i \neq i' \in A, \\ W_j(x^\eta) &\geq J_{j \leftarrow j'}^B(x^\eta) - \eta & \forall j \neq j' \in B. \end{aligned}$$

The source obtains this by assigning every one of the $2n$ agents an equal money endowment $\eta/(2n)$ in a double-indexed-price equilibrium. The resulting allocation is Pareto optimal, and the total residual money is at most η , which yields the displayed additive inequalities.

The only additional fact needed to pass from Proposition 3.1 to exact JEF+PO is that Pareto optimality is closed in this polyhedral model. We prove that fact next.

4 Pareto-optimal points form a closed set

We state the result for a general polytope. Let $P \subseteq \mathbb{R}^N$ be a nonempty polytope and let $F : P \rightarrow \mathbb{R}^d$ be linear. A point $x \in P$ is F -Pareto optimal if no $y \in P$ satisfies $F(y) \geq F(x)$ coordinatewise with at least one strict inequality.

Lemma 4.1 (Strictly positive scalarization). *A point $x \in P$ is F -Pareto optimal if and only if there exists $\alpha \in \mathbb{R}_{++}^d$ such that*

$$x \in \operatorname{argmax}_{y \in P} \alpha^\top F(y).$$

Proof. The reverse implication is immediate: a Pareto dominator would strictly increase the objective because every coordinate of α is positive.

For the forward implication, write $P = \{y : Ay \leq b\}$. Fix an F -Pareto-optimal point x and consider the linear program

$$\begin{aligned} &\text{maximize} && \mathbf{1}^\top F(y) \\ &\text{subject to} && F(y) \geq F(x), \\ &&& Ay \leq b. \end{aligned} \tag{1}$$

The point x is feasible. Pareto optimality implies that every feasible y has $F(y) = F(x)$, so the optimum is $\mathbf{1}^\top F(x)$.

Associate dual variables $\lambda \geq 0$ with the inequalities $-F(y) \leq -F(x)$ and dual variables $\mu \geq 0$ with $Ay \leq b$. The dual is

$$\begin{aligned} & \text{minimize} && -\lambda^\top F(x) + \mu^\top b \\ & \text{subject to} && -F^\top \lambda + A^\top \mu = F^\top \mathbf{1}, \\ & && \lambda, \mu \geq 0. \end{aligned}$$

By strong duality, an optimal pair (λ, μ) satisfies

$$\mu^\top b - \lambda^\top F(x) = \mathbf{1}^\top F(x).$$

Set $\alpha := \mathbf{1} + \lambda \in \mathbb{R}_{++}^d$. Dual feasibility gives $A^\top \mu = F^\top \alpha$, and strong duality gives $\mu^\top b = \alpha^\top F(x)$. Therefore, for every $y \in P$,

$$\alpha^\top F(y) = \mu^\top Ay \leq \mu^\top b = \alpha^\top F(x).$$

Thus x maximizes the strictly positive weighted sum. $\square$

Theorem 4.2 (Polyhedral closedness of Pareto optimality). *The set of F -Pareto-optimal points of P is a finite union of faces of P . In particular, it is closed and compact.*

Proof. For each $\alpha \in \mathbb{R}_{++}^d$, the maximizer set

$$\operatorname{argmax}_{y \in P} \alpha^\top F(y)$$

is a face of P . By Lemma 4.1, the Pareto-optimal set is the union of these maximizer faces. A polytope has only finitely many faces, so this union contains only finitely many distinct faces. Each face is closed, and the union is therefore closed. It is compact because it is a closed subset of the compact set P . $\square$

Remark 4.3. The polyhedral hypothesis matters. For arbitrary compact feasible sets, the strong Pareto set need not be closed. The matching model avoids this issue because the feasible set is the Birkhoff polytope and all utilities are linear.

5 Exact JEF and Pareto optimality coexist

Theorem 5.1 (Exact coexistence). *Every finite two-sided cardinal-utility matching instance admits a fractional perfect matching that is simultaneously justified-envy-free and Pareto optimal.*

Proof. Choose a sequence $\eta_k \downarrow 0$. By Proposition 3.1, for each k there is a Pareto-optimal $x^k \in \mathcal{X}$ whose additive JEF error is at most η_k .

The Birkhoff polytope $\mathcal{X}$ is compact, so a subsequence converges to some $x^* \in \mathcal{X}$. Apply Theorem 4.2 to the $2n$ linear utility coordinates

$$F(x) = ((U_i(x))_{i \in A}, (W_j(x))_{j \in B}).$$

Because every x^k is Pareto optimal and the Pareto-optimal set is closed, x^* is Pareto optimal.

Every JEF benchmark is linear in x . Passing to the limit in the additive inequalities gives

$$U_i(x^*) \geq J_{i \leftarrow i'}^A(x^*), \quad W_j(x^*) \geq J_{j \leftarrow j'}^B(x^*)$$

for all same-side pairs. Hence x^* is JEF. $\square$

Theorem 5.1 does not require strict preferences or strictly positive utilities. In particular, allowing zero utilities cannot rescue a nonexistence construction.

Theorem 5.2 (Rational exact solution). *If all utilities are rational, then there exists a rational allocation that is simultaneously JEF and Pareto optimal. Moreover, some vertex of the JEF polytope is globally Pareto optimal.*

Proof. Let $\mathcal{P}_{\text{JEF}} \subseteq \mathcal{X}$ be the set of JEF allocations. It is a rational polytope because all JEF inequalities are linear with rational coefficients. By Theorem 5.1, choose $x^* \in \mathcal{P}_{\text{JEF}}$ that is Pareto optimal. By Lemma 4.1, there is $\alpha \in \mathbb{R}_{++}^{2n}$ such that x^* maximizes the corresponding weighted sum of all agents' utilities over $\mathcal{X}$.

Now maximize the same objective over $\mathcal{P}_{\text{JEF}}$. Since $x^* \in \mathcal{P}_{\text{JEF}} \subseteq \mathcal{X}$, the optimum over $\mathcal{P}_{\text{JEF}}$ equals the optimum over $\mathcal{X}$. Consequently, every optimizer over $\mathcal{P}_{\text{JEF}}$ is also a global optimizer over $\mathcal{X}$, and is therefore Pareto optimal. The optimizer set is a nonempty face of the rational polytope $\mathcal{P}_{\text{JEF}}$, so it contains a vertex of $\mathcal{P}_{\text{JEF}}$. Every vertex of a rational polytope is rational. $\square$

Corollary 5.3 (No multiplicative nonexistence example). *For every $C \geq 1$ and every instance, there exists a Pareto-optimal C -JEF allocation. In fact, one may take $C = 1$.*

Thus a negative resolution of the open problem cannot proceed by showing that some instance lacks a Pareto-optimal constant-factor JEF allocation. Any negative resolution would have to be computational, for example by proving hardness of finding such an allocation.

6 Why the proposed selector lemma is impossible

The submitted construction asserted that for each $m \geq 3$ and $0 < \varepsilon < 1/10$, every Pareto-optimal allocation has the form $x(p)$ for some $p \in \Delta_m$, and for each coordinate r ,

$$U_{e_r}(x(p)) = \varepsilon + (1 - \varepsilon)p_r, \quad (2)$$

$$J_{e_r \leftarrow c}^A(x(p)) \geq (1 - 2\varepsilon)(1 - p_r). \quad (3)$$

These assertions alone are inconsistent with Theorem 5.1; no analysis of the proposed locking gadgets is needed.

Proposition 6.1 (Uniform contradiction to the selector). *For $m \geq 3$ and $0 < \varepsilon < 1/10$, every $p \in \Delta_m$ has a coordinate r for which the right-hand side of (3) exceeds the quantity in (2) by more than $2/15$.*

Proof. Choose r with $p_r \leq 1/m$. Then

$$\begin{aligned} J_{e_r \leftarrow c}^A(x(p)) - U_{e_r}(x(p)) &\geq (1 - 2\varepsilon) \left(1 - \frac{1}{m}\right) - \left(\varepsilon + \frac{1 - \varepsilon}{m}\right) \\ &= 1 - \frac{2}{m} - 3\varepsilon \left(1 - \frac{1}{m}\right). \end{aligned}$$

As a function of $q = 1/m \in (0, 1/3]$, the last expression is $(1 - 3\varepsilon) + (3\varepsilon - 2)q$. Since $3\varepsilon - 2 < 0$, its minimum occurs at $q = 1/3$, where it equals $1/3 - 2\varepsilon > 2/15$. $\square$

If all Pareto-optimal allocations belonged to this selector family, every one of them would violate JEF by a fixed positive additive amount. This contradicts both Proposition 3.1 and the exact existence result in Theorem 5.1. The claimed state table therefore cannot be the Pareto frontier of any finite two-sided cardinal matching instance.

7 A polynomial-time dimension-dependent approximation

Exact existence does not by itself identify the correct Pareto face in polynomial time. We next record a simple algorithmic guarantee that is weaker than the desired constant factor but is fully constructive.

For each $i \in A$ and $j \in B$, define

$$M_i := \max_{h \in B} u_{ih}, \quad N_j := \max_{h \in A} w_{jh}.$$

First solve the max–min linear program

$$\begin{aligned} & \text{maximize} && t \\ & \text{subject to} && x \in \mathcal{X}, \\ & && U_i(x) \geq tM_i \quad \forall i \in A, \\ & && W_j(x) \geq tN_j \quad \forall j \in B, \\ & && 0 \leq t \leq 1. \end{aligned} \tag{4}$$

Let (x^0, t^*) be an optimal solution. Next solve the Pareto-refinement linear program

$$\begin{aligned} & \text{maximize} && \sum_{i \in A} U_i(y) + \sum_{j \in B} W_j(y) \\ & \text{subject to} && y \in \mathcal{X}, \\ & && U_i(y) \geq U_i(x^0) \quad \forall i \in A, \\ & && W_j(y) \geq W_j(x^0) \quad \forall j \in B. \end{aligned} \tag{5}$$

Theorem 7.1 (Polynomial-time $1/t^*$ approximation). *Every optimizer y^* of (5) is Pareto optimal and $1/t^*$ -JEF. Moreover, $t^* \geq 1/n$, so the returned allocation is always n -JEF. Both linear programs have polynomial size and can be solved in polynomial time.*

Proof. Suppose $z \in \mathcal{X}$ Pareto dominates y^* . Because y^* weakly improves every agent relative to x^0 , so does z . Thus z is feasible for (5), and its objective value is strictly larger than that of y^* , a contradiction. Hence y^* is Pareto optimal.

Fix distinct $i, i' \in A$. Since the i' th row of y^* has total mass one,

$$J_{i \leftarrow i'}^A(y^*) \leq M_i \sum_{j \in B: w_{ji} \geq w_{ji'}} y_{ij}^* \leq M_i.$$

By feasibility of (4) and (5),

$$U_i(y^*) \geq U_i(x^0) \geq t^* M_i.$$

Therefore $J_{i \leftarrow i'}^A(y^*) \leq U_i(y^*)/t^*$. The proof for agents on side B is identical.

Finally, the uniform allocation $\bar{x}_{ij} = 1/n$ is feasible for $t = 1/n$, because

$$U_i(\bar{x}) = \frac{1}{n} \sum_{j \in B} u_{ij} \geq \frac{M_i}{n}, \quad W_j(\bar{x}) = \frac{1}{n} \sum_{i \in A} w_{ji} \geq \frac{N_j}{n}.$$

Hence $t^* \geq 1/n$. □

The value $1/t^*$ is an efficiently computable instance-dependent factor. For example, the algorithm gives a constant guarantee on any class of instances for which all agents can simultaneously obtain a constant fraction of their individual maximum utility. In the worst case, however, $t^* = \Theta(1/n)$ is possible, so this argument does not resolve the constant-factor open problem.

8 What remains open

Theorem 5.1 shows that exact JEF and Pareto optimality are always compatible. The remaining difficulty is to find the appropriate Pareto face efficiently. Equivalently, one seeks $x \in \mathcal{P}_{\text{JEF}}$ and a strictly positive vector α such that

$$x \in \operatorname{argmax}_{y \in \mathcal{X}} \alpha^\top ((U_i(y))_{i \in A}, (W_j(y))_{j \in B}).$$

For a fixed α , this is a linear program, but the supporting vector is not known in advance. Theorem 5.2 gives a finite exact algorithm by enumerating vertices of the JEF polytope and testing each vertex for global Pareto optimality with a linear program. The number of vertices can be exponential, so this is not a polynomial-time solution.

Accordingly, the correct status of Open Problem W4414541699 is:

Exact JEF+PO allocations always exist, even rationally. Linear programming gives an approximation factor of n in polynomial time. It remains open whether a dimension-free constant factor can be achieved in polynomial time.

9 Conclusion

The attempted negative construction cannot be repaired as an existential counterexample. The additive Pareto-optimal approximants already established by Tröbst and Vazirani converge, after passing to a subsequence, to an exact JEF allocation; polyhedral closedness ensures that the limit remains Pareto optimal. This yields exact and rational coexistence for every instance. The substantive unresolved issue is computational rather than existential.

References

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